

NJIT Final Round Interview: Technical Case Study

Position: AI Analytics Engineer

Department: Analytics and Data Governance (ADG) + Interview Committee

Format: 15-Minute Presentation (or less) + Q&A

Overview & Expectations

This session focuses on your data architecture strategy and your ability to communicate complex technical concepts to a diverse audience.

- **The Audience:** You will present to a cross-functional search committee that includes both technical engineers and non-technical university leaders. **You are expected to present your strategy in layman's terms**, demonstrating how you translate complex backend data engineering into clear institutional value.

The Scenario

You are provided with a Synthetic 10-year dataset (2016–2026) tracking student records across Fall and Spring semesters. The dataset contains demographics (gender, Pell Grant status), academic progress (degree levels, programs, semester registration status), and outcomes (graduation status/term).

Your Challenge

Choose **ONE** of the following application pathways and explain how you would engineer the data foundation to power it safely and reliably. Regardless of the option you choose, your underlying data structure must be designed to accurately answer complex institutional questions such as:

- *How have graduation rates for Pell-eligible students shifted over the last decade?*
- *What is the first-year retention rate for undergraduate students in the College of Engineering?*
- *What is the number of Pell Grant-eligible students by semester in Master's level programs?*

Select Your Delivery Option:

- **Option A: AI Conversational Chatbot** – A natural language interface allowing campus executives to type these questions directly and receive immediate, accurate text responses.
- **Option B: Interactive Self-Service Dashboard** – A dashboard application built in a framework of your choice (e.g., Streamlit, Dash) or a traditional BI platform (e.g.,

Tableau). This application answers these questions using precalculated metrics, logic, and user-driven filters.

Appendix: Dataset Schema & Data Dictionary

The provided dataset contains longitudinal, semester-by-semester student records spanning 2016 to 2026. Use the following data dictionary to map the fields and codes for your data architecture design.

field Name	Data Type	Description & Valid Values
student_id	Text/Numeric	A unique, masked identifier for each individual student.
u_g	Character	Student Level Indicator: • U = Undergraduate • G = Graduate (Master's level) • D = Doctoral
gender	Character	Student Gender: Female , Male .
school	Character	Student School: AD, CC, EN, SL, SM
degtype	Text	Degree Type: The specific degree program abbreviation (e.g., BA, BS, MS, PhD).
firstgen	Character	First-Generation Student Indicator: • Y = First-generation college student • N = Not a first-generation college student
pell	Character	Pell Grant Eligibility Indicator: • Y = Pell Grant eligible (Low-income baseline) • N = Not Pell Grant eligible
ethnicmultirace	Text	Demographic Race/Ethnicity: Category describing the student's reported racial or ethnic background.
citizen	Text/Number	Citizenship Status Category describing the student's reported citizenship status

creditnr	Numeric	Enrolled Credits: Total credit hours the student is registered for in the given semester.
accumgpa	Decimal	Cumulative GPA: The student's cumulative Grade Point Average at the <i>beginning</i> of the corresponding semester.
term_code	Text/Number	Term Code: The term code corresponding to the enrollment record

Code Mapping Tables

1. Academic Units (**school**)

Represents the primary college or school hosting the student's academic major.

Code	Institutional Unit Description
AD	Hillier College of Architecture and Design (HCAD)
CC	Ying Wu College of Computing (YWCC)
EN	Newark College of Engineering (NCE)
SL	College of Science and Liberal Arts (CSLA)
SM	Martin Tuchman School of Management (MTSM)

2. Citizenship Status (**citizen**)

Code	Status Description
1	US Citizen
2	International
3	US Permanent Resident
4	Information Not

	Available
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3. Registration Status (**regstat**)

Defines the student's enrollment type for the given semester.

Cod e	Status Category	Description
1	First-time	New students entering college for the very first time.
2	Transfer	Students entering NJIT with credits from another institution.
3	Stop-out / Readmit	Returning students re-enrolling after an unapproved absence.
4	Continuing	Active, ongoing students progressing through their degrees.

4. Academic Terms (**term_code**)

Each instance of an academic term is represented by a six-digit code in the format **YYYYTT** (e.g., 202490).

- The first four digits (**YYYY**) represent the active calendar year.
- The last two digits (**TT**) represent the specific academic term category:

Term Suffix (TT)	Academic Term Description
10	Spring Semester
50	Summer Term
90	Fall Semester
95	Winter Intersession

5. Ethnicity / Race Categories (**ethnicmultirace**)

Code	Category Description
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1	Native American-Alaskan Native
2	Aboriginal
3	Central African (Congo, Kenya, Uganda, etc.)
4	Southern African (Namibia, Madagascar, etc.)
5	Northern African (Morocco, Algeria, etc.)
6	African American/Black
7	Caribbean (Haiti, Dominican Republic, etc.)
8	Central American (Nicaragua, Guatemala, etc.)
9	Cuban
10	Latino
11	Hispanic
12	Mexican
13	Puerto Rican
14	South American (Brazil, Venezuela, Argentina, etc.)
15	Central Asian (Afghanistan, Kazakhstan, etc.)
16	Southeast Asian (Indonesia, Thailand, Vietnam, etc.)
17	South Asian (India, Pakistan, Bangladesh, etc.)
18	East Asian (China, Japan, Korea, Philippines, etc.)
19	Pacific-Islander (Fiji, Tahiti, Hawaii, etc.)
20	Asian American
21	Iberian-European (Spanish-Portuguese)
22	Eastern European (Ukraine, Poland, Albania, etc.)
23	Indo-European (Turkey, Azerbaijan, etc.)

24	European (Italy, Sweden, Germany, etc.)
25	Middle Eastern (Israel, Jordan, Lebanon, etc.)
26	White American
27	Australian/New Zealander